

Figure 1.36

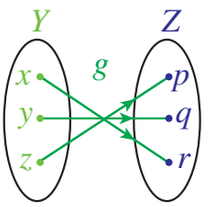


Figure 1.37

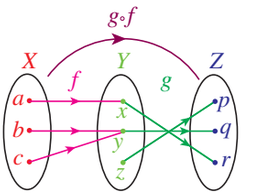


Figure 1.38

Now we can define a function $h : X \to Z$ in a natural way. To find the image of a under h, we first see the image of a under f; it is x; then we see the image of this x under g; this is r. That is, $h(a) = r$. Similarly, we declare $h(b) = q$ and $h(c) = q$. In this way we can define a new function h. This h is called the composition of f with g.

Definition 1.8

Let $f : X \to Y$ and $g : Y \to Z$ be two functions. Then the function $h : X \to Z$ defined as $h(x) = g(f(x))$ for every $x \in X$ is called the ***composition*** of f with g. It is denoted by $g \circ f$ (Read this as f composite with g). (See Figures 1.38 and 1.39.)

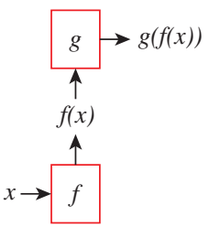


Figure 1.39

We can note that the range of f need not be Y. If $f : X \to Y_1, g : Y_2 \to Z$ and $Y_1 \subseteq Y_2$, then also we can define $g \circ f$; we can take Y_2 as the co-domain of f and use the same definition. So we can define $g \circ f$ if and only if the range of f is contained in the domain of g.

Example 1.25 Let $f = \{(1, 2), (3, 4), (2, 2)\}$ and $g = \{(2, 1), (3, 1), (4, 2)\}$. Find $g \circ f$ and $f \circ g$.

Solution:

To check whether compositions can be defined, let us find the domain and range of these functions.

Domain of $f = \{1, 2, 3\}$, Range of $f = \{2, 4\}$, Domain of $g = \{2, 3, 4\}$ and Range of $g = \{1, 2\}$. Since the range of f is contained in the domain of g we can define $g \circ f$; so as to find the image of 1 under $g \circ f$, we first find the image of 1 under f and then its image under g. The image of 1 under f is 2 and its image under g is 1. So $(g \circ f)(1) = g(f(1)) = g(2) = 1$.